

Boundary-Safe Global Synthesis for the Dubins Car

Route-A source-local certificate HCS-C310

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Abstract

We give a complete executable form of the Dubins shortest-path theorem: all six *CSC/CCC* words, feasibility tests, segment triples, and the global all-ties minimum for arbitrary oriented planar endpoints and turning radius. The theorem closes collapsed pieces, tangent and three-circle boundaries, coincident poses, inverse-angle conventions, scaling, reflection, and direct endpoint replay.

1 Six-word global synthesis

At unit speed and turning radius $R > 0$, the car satisfies

$$\dot{x} = \cos \vartheta, \quad \dot{y} = \sin \vartheta, \quad |\dot{\vartheta}| \leq R^{-1}, \quad (1)$$

and moves only forward. Rigid motion fixes the initial pose at $(0, 0, 0)$. Divide the target position by R and write

$$d = R^{-1} \sqrt{x^2 + y^2}, \quad \theta = \text{atan2}(y, x), \quad \alpha = \text{mod}_{2\pi}(-\theta), \quad \beta = \text{mod}_{2\pi}(\phi - \theta).$$

Here $\text{mod}_{2\pi} z$ is the representative in $[0, 2\pi)$, with $\text{mod}_{2\pi}(2\pi) = 0$. Put $s_a = \sin \alpha$, $c_a = \cos \alpha$ and similarly for b , and $c_{ab} = \cos(\alpha - \beta)$. The following table is an explicit algorithm. Every square root requires its radicand $P \geq 0$.

word	normalized segments (t, p, q) and feasibility
<i>LSL</i>	$P = 2 + d^2 - 2c_{ab} + 2d(s_a - s_b)$, $\psi = \text{atan2}(c_b - c_a, d + s_a - s_b)$; $(\text{mod}_{2\pi}(-\alpha + \psi), \sqrt{P}, \text{mod}_{2\pi}(\beta - \psi))$.
<i>RSR</i>	$P = 2 + d^2 - 2c_{ab} + 2d(s_b - s_a)$, $\psi = \text{atan2}(c_a - c_b, d - s_a + s_b)$; $(\text{mod}_{2\pi}(\alpha - \psi), \sqrt{P}, \text{mod}_{2\pi}(-\beta + \psi))$.
<i>LSR</i>	$P = -2 + d^2 + 2c_{ab} + 2d(s_a + s_b)$, $\psi = \text{atan2}(-c_a - c_b, d + s_a + s_b) - \text{atan2}(-2, \sqrt{P})$; $(\text{mod}_{2\pi}(-\alpha + \psi), \sqrt{P}, \text{mod}_{2\pi}(-\beta + \psi))$.
<i>RSL</i>	$P = -2 + d^2 + 2c_{ab} - 2d(s_a + s_b)$, $\psi = \text{atan2}(c_a + c_b, d - s_a - s_b) - \text{atan2}(2, \sqrt{P})$; $(\text{mod}_{2\pi}(\alpha - \psi), \sqrt{P}, \text{mod}_{2\pi}(\beta - \psi))$.
<i>RLR</i>	$h = (6 - d^2 + 2c_{ab} + 2d(s_a - s_b))/8$, $ h \leq 1$; $p = \text{mod}_{2\pi}(2\pi - \arccos h)$, $t = \text{mod}_{2\pi}(\alpha - \text{atan2}(c_a - c_b, d - s_a + s_b) + p/2)$, $q = \text{mod}_{2\pi}(\alpha - \beta - t + p)$.
<i>LRL</i>	$h = (6 - d^2 + 2c_{ab} + 2d(-s_a + s_b))/8$, $ h \leq 1$; $p = \text{mod}_{2\pi}(2\pi - \arccos h)$, $t = \text{mod}_{2\pi}(-\alpha - \text{atan2}(c_a - c_b, d + s_a - s_b) + p/2)$, $q = \text{mod}_{2\pi}(\beta - \alpha - t + p)$.

Theorem 1 (Complete Dubins minimizer). *Compute every feasible row above, allowing zero segments. Then*

$$D_R((0, 0, 0), (x, y, \phi)) = R \min_{w \in \{LSL, RSR, LSR, RSL, RLR, LRL\}} (t_w + p_w + q_w). \quad (2)$$

Every minimizing row is a global shortest curve, and every global shortest curve has one of these descriptions, up to zero-piece degeneracy.

Proof. The Pontryagin switching function makes a nonsingular extremal a concatenation of maximum-curvature left/right arcs and straight segments. The classical Dubins shortening argument removes repeated equal turns, excludes four or more essential pieces, and leaves precisely *CSC* or *CCC*, hence the six rows. External and internal circle tangencies give the four radicands; the law of cosines for the three unit circles gives the two values h . Solving the remaining oriented angles yields the table. Thus evaluating all feasible rows proves (2); taking all equal minima retains every global tie. $\square$

2 Boundaries, ties, and symmetries

For a pose (X, Y, H) , a normalized primitive of length a is

$$\begin{aligned} L_a &: (X + \sin(H + a) - \sin H, Y - \cos(H + a) + \cos H, H + a), \\ R_a &: (X + \sin H - \sin(H - a), Y + \cos(H - a) - \cos H, H - a), \\ S_a &: (X + a \cos H, Y + a \sin H, H). \end{aligned}$$

Substitution of each table row gives $(x/R, y/R, \phi)$ modulo 2π ; this is a direct endpoint proof independent of comparing lengths. It also shows that $R(t + p + q) \geq \sqrt{x^2 + y^2}$.

At $P = 0$ the straight tangent collapses. At $h = 1$ the middle *CCC* arc may collapse; at $h = -1$ it is a half turn. In these cells multiple word labels can describe the same geometric curve, and (2) deliberately keeps all of them. For deterministic boundary coordinates we set $\text{atan2}(0, 0) = 0$; changing that representative only redistributes zero or full-turn pieces and cannot lower the geometric minimum. Coincident poses, a pure heading change, heading wrap, and a completely zero path are therefore covered without an informal exception.

For every $s > 0$, simultaneous replacement $(x, y, R) \mapsto (sx, sy, sR)$ multiplies the answer by s . Rigid motions of both poses preserve it. Reflection $(x, y, \phi) \mapsto (x, -y, -\phi)$ exchanges $L \leftrightarrow R$, pairing *LSL/RSR*, *LSR/RSR*, and *LRL/RLR*.

Source lineage

References

- [1] L. E. Dubins, “On curves of minimal length with a constraint on average curvature, and with prescribed initial and terminal positions and tangents,” *American Journal of Mathematics* 79 (1957), 497–516. DOI: [10.2307/2372560](https://doi.org/10.2307/2372560).